# Joseph Breda

PhD Candidate · Paul G. Allen School of Computer Science & Engineering
joebreda.github.io · joebreda@cs.washington.edu

## Research

My impact-driven research focuses on developing and evaluating **AI** alongside **population-scale signals** to enhance **societal health and well-being**. I have designed and led large-scale evaluations of LLMs for differential diagnosis via population deployments, directed a multi-year flu challenge study with wearables to use digital biomarkers to predict pre-symptomatic onset of the flu, and built geospatial computer vision models to map traffic safety. My work addresses the practical challenges of aligning society-scale technologies with human-centric applications and real populations.

## Education

2019 - 2026 Ph.D.
**University of Washington,** *Computer Science*
Thesis: *Ubiquitous Computing Platforms as Scalable Surrogates in Public Health and Well-being (Defended 2025)*
Advisor: *Shwetak Patel*

2015 - 2019 B.S.
**University of Massachusetts Amherst**, *Electrical & Computer Engineering*
Advisor: *Jay Taneja*

## Select Projects

**SymptomAI: Conversational AI Agents for Differential Diagnosis** *Nature, In Submission*

- Leading population-scale deployment and clinical evaluation of LLMs for differential diagnosis with real patients (N=13,000), validating LLM verifier for scalable oversight.
- Bridging validated AI diagnoses with population-scale wearable data, enabling association of digital biomarkers with disease phenotypes at scale.

**Biomarkers for Flu Onset Prediction via Clinical Challenge Study** *Nature, In Preparation*

- Leading flu challenge study in collaboration with NIH/NIAID, deploying commodity wearables with patients intentionally infected with the H10N7 “bird” flu.
- Modeled pre-symptomatic immune activation with wearable biomarkers, predicting flu onset, validated against gene expression from longitudinal bloodwork.

**FeverPhone: Accessible Fever Monitoring with Unmodified Smartphones** *IMWUT, 2023*

- Developed ML model mapping smartphone temperature during user-contact to measure core-body temperature, evaluated in clinic with real patients.
- Won ***Distinguished Paper Award*** at Ubicomp 2024, Highlighted in GetMobile Magazine.

**ProxiCycle: Passively Mapping Cyclist Safety with Near-Miss Detection** *CHI, 2025*

- Developed custom embedded system for passively crowdsensing geospatial cyclist safety, validated through 2-month longitudinal deployment with 15 cyclists (1000 miles traveled).
- Received national media attention, featured on NPR All Things Considered.

**Double-Blind Deployment Study of LLMs for Triaging in Emergency Room** *Working Paper*

- Designed and ran prospective double-blind study deploying LLM chatbots in clinical waiting rooms to capture diagnostic reasoning and compare it against physician notes.
- Building mixed-methods analysis pipeline to characterize LLM failure modes and identify systematic divergence between LLM and clinical-expert reasoning turn-by-turn.

## Industry Experience

**Google** Student Researcher *July 2025-Present*

- Leading SymptomAI project, evaluating conversational AI for differential diagnosis in population-scale deployment (N=13,000).
- Fine-tuning and evaluating wearable foundation model for downstream women's health forecasting tasks, targeting improvements on underrepresented data.

**Google** Student Researcher *May 2021-Oct. 2022*

- Developed geospatial deep learning model forecasting collision rates from visual road features and historic user data.

**Google** SWE Intern *May 2020-Sept. 2020*

- Developed synthetic population generation pipeline used for evaluation of Google Maps traffic models and epidemiological models of disease spread during Covid-19 pandemic.

## Publications

P18 FlowRing: Integrated Microgesture and Surface Interaction for Versatile XR Input
Ishan Chatterjee, Jiexin Ding, Anandghan Waghmare, **Joseph Breda**, Yuquan Deng, Bo Liu, Yuntao Wang, Shwetak Patel
*MHCI 2025*

P17 Lessons Learned From FeverPhone: Towards Scalable, Accessible At-Home Diagnostics via Fever Detection on Unmodified Smartphones
**Joseph Breda**, Mastafa Springston, Alex Mariakakis, Shwetak Patel
*GetMobile 2025*

P16 MobileSense: Tracking Air Pollution Exposure of Frontline Motorcycle Riders Using a Mobile Low-Cost Sensor Platform
Raunak Mukhia, Adisorn Lertsinsrubtavee, Preechai Mekbungwan, Kalana GS Jayarathna, Nussara Tieanklin, **Joseph Breda**, Ratchaphon Samphutthanont, Thongchai Kanabkaew, Kurtis Heimerl, Kanchana Kanchanasut
*COMPASS 2025*

P15 The Air We Share: Seven-Month Mixed Method Study of Thailand's Motorcycle Rideshare Driver and Air Pollution
Tieanklin, Chanwut Kittivorawong, **Joseph Breda**, Adisorn Lertsinsrubtavee, Preechai Mekbungwan, Raunak Mukia, Kalana G.S. Jayarathna, Ratchaphon Samphutthanont, Chayanon Sawatdeenarunat, Kurtis Heimerl
*Computing and Sustainable Societies 2025*

P14 ProxiCycle: Passively Mapping Cyclist Safety Using Smart Handlebars for Near-Miss Detection
**Joseph Breda**, Keyu Chen, Thomas Ploetz, Shwetak Patel
*CHI 2025*

P13 NightLight: Passively Mapping Nighttime Sidewalk Light Data for Improved Pedestrian Routing
**Joseph Breda***, Daniel Campos Zamora*, Shwetak Patel, Jon Froehlich
*CHI 2025*

P12 Exploring and Characterizing Large Language Models for Embedded System Development and Debugging
Zachary Englhardt, Richard Li, Dilini Nissanka, Zhihan Zhang, Girish Narayanswamy, **Joseph Breda**, Xin Liu, Shwetak Patel, Vikram Iyer
*CHI Late Breaking Work 2024*

P11 'I will just have to keep driving': A Mixed-methods Investigation of Lack of Agency within the Thai Motorcycle Rideshare Driver Community
*Nussara Tieanklin, ***Joseph Breda**, Tim Althoff, Kurtis Heimerl
*CSCW 2024*

P10 Thermal Earring: Low-power Wireless Earring for Longitudinal Earlobe Temperature
Qiuyue Shirley Xue, Yujia Liu, **Joseph Breda**, Mastafa Springston, Vikram Iyer, Shwetak Patel
*IMWUT 2024*

P9 Understanding People's Concerns and Attitudes Toward Smart Cities
Pardis Emami-Naeini, **Joseph Breda**, Wei Dai, Tadayoshi Kohno, Kim Laine, Shwetak Patel, Franziska Roesner
*CHI 2024*

P8 Feverphone: Accessible Core-Body Temperature Sensing for Fever Monitoring Using Commodity Smartphones
**Joseph Breda**, Mastafa Springston, Alex Mariakakis, Shwetak Patel
*IMWUT 2023* **Won Distinguished Paper Award**

P7 SpiroMask: Measuring Lung Function Using Consumer-Grade Masks
Rishiraj Adhikary, Dhruvi Lodhavia, Chris Francis, Rohit Patil, Tanmay Srivastava, Prerna Khanna, Nipun Batra, **Joseph Breda**, Jacob Peplinski, Shwetak Patel
*ACM Transactions on Computing for Health 2023*

P6 Passively Sensing SARS-CoV-2 RNA in Public Transit Buses
Jason Hoffman, Matthew Hirano, Nuttada Panpradist, **Joseph Breda**, Parker Ruth, Yuanyi Xu, Jonathan Lester, Bichlien H. Nguyen, Luis Ceze, Shwetak Patel
*Science of the Total Environment 2022*

P5 Phone-based Ambient Temperature Sensing Using Opportunistic Crowdsensing and Machine Learning
Amee Trivedi, Phuthipong Bovornkeeratiroj, **Joseph Breda**, Prashant Shenoy, Jay Taneja
*Sustainable Computing 2021*

P4 Hanging Gardens of Babylon: Reframing Urban Agriculture as an Opportunity for Social Engagement
**Joseph Breda**, Esther Jang, Kurtis Heimerl, Shwetak Patel
*Self-Sustainable CHI 2020*

P3 Hot or Not: Leveraging Mobile Devices for Ubiquitous Temperature Sensing.
**Joseph Breda**, Amee Trivedi, Chulabhaya Wijesundara, Phuthipong Bovornkeeratiroj, David Irwin, Prashant Shenoy, Jay Taneja
*BuildSys 2019*

P2 Staring at the Sun: A Physical Black-box Solar Performance Model
Dong Chen, **Joseph Breda**, David Irwin
*BuildSys 2018*

P1 Fancy That: Measuring Electricity Grid Voltage Using a Phone and a Fan.
**Joseph Breda** and Jay Taneja
*COMPASS 2018*

## Organizing Experience

O2 **Founded and President of the Allen School Graduate Entrepreneurship Club** 2023-Present
Organizing quarterly panels and event connecting academic start up founders with current PhD students interested entrepreneurship.

O1 **Founding member of CS4Env cross-department collaborative initiative.** 2022-2023
Assisted with early organizing and developed website during first year launch.

## Teaching Experience

T3 **Embedded Systems Capstone Teaching Assistant** Winter 2024
See [T2].

T2 **Embedded Systems Capstone Teaching Assistant** Spring 2024
Mentored teams of students on end-to-end capstone projects and led lectures on embedded ML and Android BLE.

T1 **Embedded Systems Teaching Assistant** Fall 2023
Tutored embedded systems during office hours and graded assignments

## Academic Service

AS7 **UW CSE PhD Area Chair** 2025
Triaged over 30 PhD Applications and served as area chair in support of Ubicomp and HCI faculty decisions on PhD Admissions

AS6 **Paper Reviewer** 2025
Reviewed for ACM IMWUT, ACM CHI, ACM UIST & ACM COMPASS

AS5 **Paper Reviewer** 2024
Reviewed for ACM IMWUT & ACM CHI

AS4 **High school Intern Outreach and Mentor** 2024
Interviewed and mentored 1 team of 5 high school summer lab interns who built a cognitive testing game.

AS3 **Paper Reviewer** 2023
Reviewed for ACM CSCW

AS2 **High school Intern Outreach and Mentor** 2023
Interviewed and mentored 2 teams of 4 high school summer lab interns who built a zoom-call AI summarization platform.

AS1 **UW CSE PhD Admission Reader** 2022
Triaged over 60 PhD Applications

## Awards

A10 **Samsung Global Research Outreach (GRO) Grant** November 2024
Primary author on **$150,000** grant proposal extending my prior work collaborating with the NIH for flu monitoring and early detection.

A9 **Distinguished Paper Award at Ubicomp 2024** October 2024
For work on my first author paper [P8].

A8 **AI-Focused Population Health Pilot Grant** June 2024
Primary contributor to **$100,000** university level pilot grant to for population health applications of AI

A7 **Young Researcher** October 2023
Invited and attended 10th Heidelberg Laureate Forum

A6 **Computing for the Environment Initiative Grant** June 2022
Primary author on 2 project proposals totaling **$100,000** of funding ($50,000 each) for developing computer systems for sustainability.

A5 **Google Gift Grant** October 2021
Primary author of **$60,000** grant to study human mobility patterns.

A4 **Weil Family Endowed Fellowship in Computer Science & Engineering** September 2019
Selected for award upon PhD admission.

A3 **Graduated from Commonwealth Honors College** May 2019
For completing honors undergraduate thesis, later published as [P3].

A2 **Graduated Magna Cum Laude** May 2019
Top 10% of graduating class within the ECE department.

A1 **Commonwealth Honors College: Honors Research Grant** December 2018
Awarded research funding for proposed honors thesis.

## Select Press

PR3 **NPR All Things Considered:** This week in science: biker safety, orange cats and a gum disease-heart rhythm link 2025

PR2 **GeekWire:** UW develops bike-mounted sensor system to track close-passing cars and map safer cycling routes 2025

PR1 **Gizmodo**: Researchers Created a Simple App That Turns Your Smartphone Screen Into an Accurate Thermometer 2023

## Select Skills

**Machine Learning & AI**: AI Evaluation Frameworks | LLM-as-Judge / Model Verifiers | Foundation Model Fine-Tuning | Multimodal Learning | Computer Vision | Time-Series Modeling | Signal Processing | Geospatial ML | Deep Learning | Failure Mode Analysis | Human-AI Interaction Studies

**Research Methodology**: Empirical Analysis | Experimental Design | Ablation & Robustness Testing | Randomized Prospective Study Design | Clinical Validation | Expert Annotation Study | Population Deployment | Pre-Registration & IRB Protocols | Quantitative & Qualitative Methods | Statistical Analysis & Causal Inference

**Languages & Frameworks**: Python (Primary) | Pandas | Numpy | PyTorch | Tensorflow | Scikit-Learn | C / C++ | Android Development

**Engineering**: End-to-End ML Pipelines | Multi-Agent Systems | ETL Pipelines | System Scaling | Embedded Systems | Hardware Prototyping | Computational Fabrication | Real-World Data Pipelines

**Communication & Collaboration**: Pair Programming | Scientific Writing & Public Information Dissemination | Cross-Disciplinary Collaboration | Translational Research | Mentorship | Grant Writing & Project Proposal